LESSON 6.13

Prime Factorization

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write a number as a product of its prime factors using a factor tree.

WHY THIS MATTERS The space station received 60 supply crates. Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods. Help the crew find the prime factorization of 60!

REMEMBER

Prime Factorization

Prime Factorization & Factor Trees Prime factorization means writing a number as a product of prime numbers only. A prime number can be divided only by 1 and itself, like 2, 3, 5, and 7.

1. Prime Number: Has exactly two factors (1 and itself, e.g. 2, 3, 5, 7, 11).
2. Split composite number into factor pairs until every branch ends in a prime.
3. Write final product using exponents in ascending order (e.g. 60 = $2^2 \times 3 \times 5$)

Words: **Prime Factorization** ***(Factorización prima)*** **Prime number** ***(Número primo)*** **Composite number** ***(Número compuesto)*** **Factor** ***(Factor)*** **Factor tree** ***(Árbol de factores)***

1 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 6 and 9?

(A) 3
(B) 18
(C) 27
(D) 54

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

List the first four multiples of 7.

(A) 1, 7, 14, 21

(B) 7, 14, 21, 28

(C) 7, 7, 7, 7

(D) 7, 17, 27, 37

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 5 and 10?

(A) 5

(B) 10

(C) 15

(D) 50

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Hot dogs come in packs of 10 and buns in packs of 8. What is the smallest number of each you can buy with none left over?

(A) 18

(B) 20

(C) 40

(D) 80

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A station gardener is loading 72 seed pods into the greenhouse and wants to build a perfectly rectangular planting grid. Starting from the prime factorization of 72, she combines the prime factors to list every factor of 72 and choose a grid that uses all the pods. What is the prime factorization of 72?

(A) $2^3 \times 3^2$
(B) $2^2 \times 3^3$
(C) 6×12
(D) 8×9

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A station gardener is loading 72 seed pods into the greenhouse and wants to build a perfectly rectangular planting grid. Starting from the prime factorization of 72, she combines the prime factors to list every factor of 72 and choose a grid that uses all the pods. The gardener wants a grid with 8 rows. How many pods in each row so all 72 are used?

(A) 6
(B) 9
(C) 12
(D) 64

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

SENTENCE STARTER I know ___ is prime because its only factors are ___ and ___. I know ___ is composite because it has factors like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can write a number as a product of its prime factors using a factor tree.

WHY THIS MATTERS The space station received 60 supply crates. Mission Control needs to break this quantity into its prime components so the sorting robots can distribute them into equally sized pods. Help the crew find the prime factorization of 60!

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort the station ID numbers: drag each one into PRIME (only 1 and itself) or COMPOSITE (more than two factors) to clear the scanner.

Groups: Prime Numbers | Composite Numbers

_____ 19

_____ 37

_____ 41

_____ 63

_____ 44

_____ 99

2 **Write the letter of the matching item on each line.**

Match each access number to its prime factorization to power up the airlock.

_____ 42

_____ 70

_____ 75

_____ 66

_____ 88

_____ 63

(A) $2 \times 3 \times 7$

(B) $2 \times 5 \times 7$

(C) $3 \times 5 \times 5$

(D) $2 \times 3 \times 11$

(E) $2 \times 2 \times 2 \times 11$

(F) $3 \times 3 \times 7$

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A station gardener is loading 72 seed pods into the greenhouse and wants to build a perfectly rectangular planting grid. Starting from the prime factorization of 72, she combines the prime factors to list every factor of 72 and choose a grid that uses all the pods. Which grid would NOT use exactly all 72 seed pods?

(A) 3 by 24

(B) 4 by 18

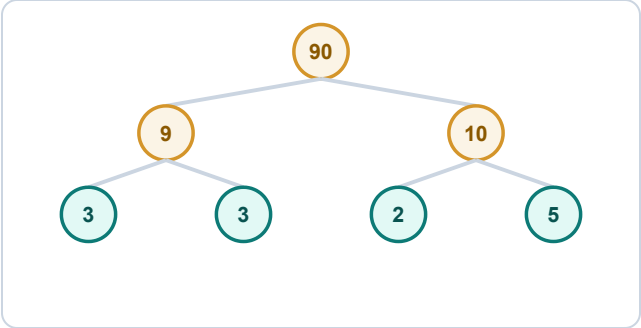
(C) 5 by 14

(D) 6 by 12

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Factor Tree for 90



The first lock panel shows 90. What is the prime factorization of 90?

(A) 2 × 3 × 3 × 5

(B) 2 × 45

(C) 3 × 30

(D) 9 × 10

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

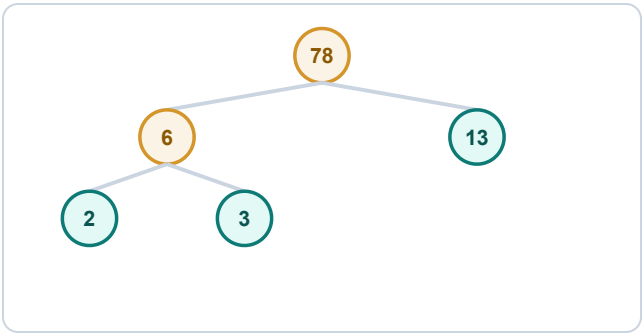
The second panel needs a number whose prime factorization is $2^2 \times 3^2$. Which number unlocks it?

(A) 12
(B) 18
(C) 24
(D) 36

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Factor Tree for 78



The final lock asks for the prime factorization of 78 written as a product of primes. Which is correct?

(A) $2 \times 3 \times 13$
(B) 2×39
(C) 3×26
(D) 6×13

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide whether a number is prime or composite? What strategy did you use?

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